

Text Classification: Definitions, Representations, Models, and Evaluation

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

classification maps data items to predefined finite classes → types include binary, SLMC, and MLMC → rule-based methods trade interpretability for coverage → supervised pipeline labels data, extracts features, trains classifier → feature extraction turns text into vectors → feature selection/weighting controls sparsity and importance → classifiers must handle high-dimensional sparse text → embeddings and PLMs provide dense/contextual representations → zero/few-shot use language models without task-specific fine-tuning → evaluation needs metrics, baselines, data split, and hyperparameter discipline

Text Classification

- **Text classification** is the process of classifying documents into predefined categories based on their content.
- Input: Text (document, article, sentence)
- Task: Classify into predefined one/multiple categories
- Categories:
 - Binary: relevant/irrelevant, spam .. etc.
 - Few: sports/politics/comedy/technology
 - Hierarchical: patents

Björn Ross, TTDS 2025/2026



5

Classification is and is not

- **Classification** (a.k.a. "categorization"): a common technology in data science; studied within pattern recognition, statistics, and machine learning.
- Definition: the activity of **predicting** to which among a **predefined finite set** of groups ("classes", or "categories") a data item belongs to
- Formulated as the task of generating a hypothesis (or "classifier", or "model")

$$h : D \rightarrow C$$

where $D = \{x_1, x_2, \dots\}$ is a domain of data items and
 $C = \{c_1, \dots, c_n\}$ is a finite set of classes (the **classification scheme**)

Björn Ross, TTDS 2025/2026



6

3

原 PPT 第 3 页

这节课一句话

这节课一句话：text classification 是把文本变成可学习的表示，再用规则、监督学习或预训练语言模型，把每个文本映射到预定义标签，并用合适指标检验泛化。

1. 1. Definition: 分类不是“发现类别”，而是预测预定义类别

11/12/25

Text Classification

- **Text classification** is the process of classifying documents into predefined categories based on their content.
- Input: Text (document, article, sentence)
- Task: Classify into predefined one/multiple categories
- Categories:
 - Binary: relevant/irrelevant, spam .. etc.
 - Few: sports/politics/comedy/technology
 - Hierarchical: patents

Björn Ross, TTDS 2025/2026



5

Classification is and is not

- **Classification** (a.k.a. “categorization”): a common technology in data science; studied within pattern recognition, statistics, and machine learning.
- Definition: the activity of **predicting** to which among a **predefined finite** set of groups (“classes”, or “categories”) a data item belongs to
- Formulated as the task of generating a hypothesis (or “classifier”, or “model”)

$$h : D \rightarrow C$$

where $D = \{x_1, x_2, \dots\}$ is a domain of data items and
 $C = \{c_1, \dots, c_n\}$ is a finite set of classes (the **classification scheme**)

Björn Ross, TTDS 2025/2026



6

3

原 PPT 第 3 页

人话直觉

classification 像把邮件投进已经写好标签的信箱：spam/not spam、sports/politics、positive/negative。信箱提前存在，模型学的是投递规则。

精确定义 / 机制： Text classification 是根据文本内容把 documents、articles、sentences、queries 或 web pages 分类到 predefined categories。一般分类可表示为生成 hypothesis/classifier/model h ，把 domain D 中的数据项映射到有限 class set C 。它不同于 clustering：clustering 的 groups 和数量事先未知。若类别成员可确定性计算，例如判断自然数是否 prime，也不适合作为普通 classification 任务。

$$h : D \rightarrow C, \quad D = \{x_1, x_2, \dots\}, \quad C = \{c_1, \dots, c_n\}$$

考试问法：定义题要包含 predefined finite classes；比较题要说 classification classes known in advance, clustering groups unknown。

2. 2. Types: binary、SLMC、MLMC 看的是标签集合结构

11/12/25

Types of Classification

- **Binary:**
item to be classified into one of two classes
 $h : D \rightarrow C, C = \{c_1, c_2\}$
 - e.g., Spam/not spam, offensive/not offensive, rel/irrel
- **Single-Label Multi-Class (SLMC)**
item to be classified into only one of n possible classes.
 $h : D \rightarrow C, C = \{c_1, \dots, c_n\}$, where $n > 2$
 - e.g., Sports/politics/entertainment, positive/negative/neutral
- **Multi-Label Multi-Class (MLMC)**
item to be classified into none, one, two, or more classes
 $h : D \rightarrow 2^C, C = \{c_1, \dots, c_n\}$, where $n > 1$
 - e.g., Assigning CS articles to classes in the ACM Classification System
 - Usually be solved as n independent binary classification problems

Björn Ross, TTDS 2025/2026

9

Dimension of Classification

- Text classification may be performed according to several dimensions ("axes") orthogonal to each other
- by **topic**; by far the most frequent case, its applications are global
- by **sentiment**; useful in market research, online reputation management, social science and political science
- by **language** (a.k.a. "language identification"); useful, e.g., in query processing within search engines
- by **genre**; e.g., AutomotiveNews vs. AutomotiveBlogs, useful in website classification and others;
- by **author** (a.k.a. "authorship attribution"), by native language ("native language identification"), or by gender; useful in forensics and cybersecurity
- by **usefulness**; e.g., product reviews
-

Björn Ross, TTDS 2025/2026

10

5

原 PPT 第 5 页

人话直觉

有的任务像二选一开关，有的像多选一单选题，有的像复选题——一篇 CS 论文可以同时属于 IR、ML 和 NLP。

精确定义 / 机制： Binary classification 把 item 分到两个 classes 之一，如 spam/not spam。Single-label multi-class classification 从 $n > 2$ 个 classes 中选一个，如 sports/politics/entertainment。Multi-label multi-class classification 允许 item 属于零个、一个或多个 classes，可形式化为映射到 power set 2^C ，通常可拆成 n 个 independent binary classification problems。分类还可按 topic、sentiment、language、genre、author、gender、usefulness 等 dimensions 进行。

Binary/SLMC: $h : D \rightarrow C$; MLMC: $h : D \rightarrow 2^C$

考试问法： 分类类型题要给定义、公式和例子。MLMC 的关键是 $h:D \rightarrow 2^C$ ，不是 $h:D \rightarrow C$ ；常见解法是 one binary classifier per class。

3. 3. Rule-based classification: 清楚但脆弱的手写规则

11/12/25

Rule-based classification

- An old-fashioned way to build text classifiers was via knowledge engineering, i.e., manually building classification rules
 - E.g., (Viagra or Sildenafil or Cialis) → Spam
 - E.g., (#MAGA or America great again) → support Trump
- Common type: dictionary-based classification
- Disadvantages:
 - Expensive to setup and to maintain
 - Depends on few keywords → bad coverage (recall)

Björn Ross, TTDS 2025/2026



13

Supervised-learning classification

- A generic (task-independent) learning algorithm is used to train a classifier from a set of manually classified examples
- The classifier learns, from these training examples, the characteristics a new text should have in order to be assigned to class c
- Advantages:
 - Generating training examples cheaper than writing classification rules
 - Easy update to changing conditions (e.g., addition of new classes, deletion of existing classes, shifted meaning of existing classes, etc.)

Björn Ross, TTDS 2025/2026



14

7

原 PPT 第 7 页

人话直觉

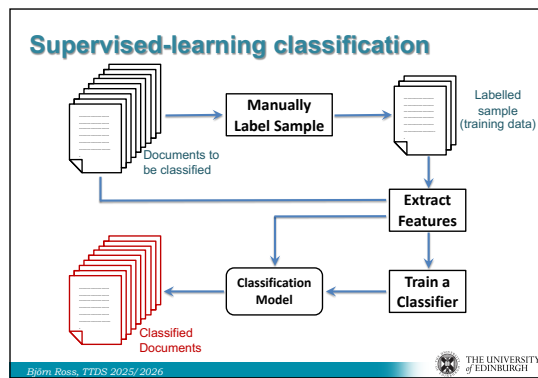
规则分类器像门卫拿着关键词清单：看到 Viagra/Cialis 就判 spam。解释很容易，但骗子换个说法就绕过去。

精确定义 / 机制： Rule-based classification 通过 knowledge engineering 手工构建规则或 dictionaries。优点是可解释、无需 labeled training data，适合规则明确或监管要求强的场景。缺点是 setup 和 maintenance 昂贵，coverage 差，依赖少量 keywords 时 recall 往往低，面对语言变化和规避策略很脆弱。

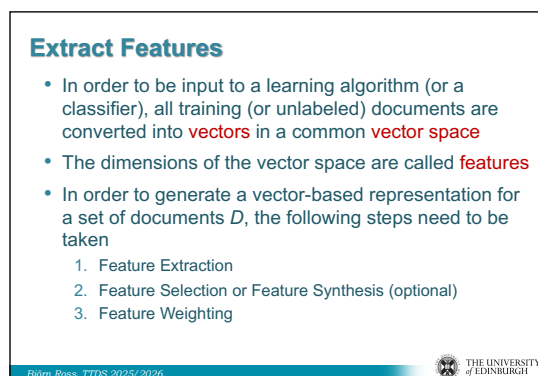
考试问法： 优缺点题答三点：interpretable/no labels vs expensive maintenance/poor coverage/low recall。举例可用 spam dictionary。

4. 4. Supervised pipeline: 让标注样本教模型，而不是人手写所有规则

11/12/25



15



16

8

原 PPT 第 8 页

人话直觉

与其让专家写一千条规则，不如先标注一批例子，让模型自己学“哪些文本特征常伴随哪个 label”。

精确定义 / 机制：Supervised-learning classification 使用人工标注样本训练 classifier。典型 pipeline 是 label sample、extract features、train classifier、apply model to unlabelled documents，并在 held-out data 上评估。它通常比构建复杂规则便宜，也更容易适应新类别和语义变化，但依赖 labeled data 的质量和代表性。

考试问法：pipeline 题要完整：labelled data → feature extraction/selection/weighting → train classifier → apply/evaluate。不要漏 evaluation。

5. 5. Feature extraction: 把文本翻译成模型能吃的向量

11/12/25

Step 1: Feature Extraction

- What are the features that should be different from one class to another?
- Simplest form: Bag-of-words (BOW)
 - Each term in a document is a feature
 - Feature space size = vocabulary in all docs
 - Standard IR preprocessing steps are usually applied
 - Tokenisation, stopping, stemming

Björn Ross, TTDS 2025/2026 

17

Step 1: Feature Extraction

- Bag-of-words (BOW)
 - Recall from Indexing lecture how we represented **documents** and words as vectors

Björn Ross, TTDS 2025/2026 

18

9

原 PPT 第 9 页

人话直觉

模型不能直接吃一篇文章，它需要一张表：每一列是 feature，每一行是 document，每个格子是这个 feature 在文档中的值。

精确定义 / 机制：大多数 learning algorithms 需要 numerical vectors。Feature extraction 决定哪些文本属性作为 vector dimensions。最简单的是 Bag-of-words：每个 term 是一个 feature，feature space size 等于训练/文档集合 vocabulary size。通常会配合 tokenisation、stopping、stemming、case folding 等 preprocessing，但选择取决于 task：sentiment 中 punctuation、all caps、emoji 可能非常有用，不应盲目删除。

$$\mathbf{x}_d = (x_{d,1}, x_{d,2}, \dots, x_{d,|V|})$$

考试问法：若问 BOW，答 each term as feature and vocabulary size as dimensionality；若问 preprocessing，强调 task-dependent。

6. 6. Beyond BOW: feature 不只有词频

11/12/25

Step 1: Feature Extraction

- What other text features could be used?
- Sentence structure:
 - POS (part-of-speech tags)
 - Syntactic tree structure
- Topic-based features:
 - LDA topics
 - NEs (named entities) in text
 - Links / Linked terms
- Non-textual features:
 - Average doc/sentence/word length
 - % of words start with upper-case letter
 - % of links/hashtags/emojis in text

Björn Ross, TTDS 2025/2026



21

Step 1: Feature Extraction

- What preprocessing to apply?
 - Case-folding? *really* vs *Really* vs *REALLY*
 - Punctuation? "?", "!", "@", "#"
 - Stopping? "he", "she", "what", "but"
 - Stemming? "replaced" vs "replacement"
- Other Features:
 - Starts with capital letter, all caps
 - Repeated characters "congraaaaats" "help!!!!!!!!!"
 - Scores from dictionaries and lexicons (e.g. LIWC)
- Which to choose?
 - Classification task/application

Björn Ross, TTDS 2025/2026



22

11

原 PPT 第 11 页

人话直觉

判断作者、情绪或网页类型时，词本身不够；句法、实体、链接、emoji、大写比例都可能泄露信号。

精确定义 / 机制：除 BOW 外，可用 word n-grams 捕捉局部词序，但会扩大 feature space 并更 sparse；character n-grams 对 OCR/ASR degraded text、拼写错误、形态丰富语言有用；syntactic features 包括 POS tags 和 syntactic tree structures；topic-based features 包括 LDA topics、named entities、links/linked terms；non-textual features 包括平均文档/句子/词长、大写比例、链接/hashtag/emoji 比例等。
考试问法：特征题不要只背 BOW。答题最好说明 feature choice depends on application，例如 sentiment 保留 emoji/all caps，language ID 可用 character n-grams。

7. 7. Feature selection: 高维稀疏是文本分类的地心引力

11/12/25

Step 2: Feature Selection

- Number of distinctive features = length of feature vector
- Vector can be of length in the order of 10^6 , and might be sparse
 - High computational cost
 - Overfitting
- What are the most important features among those?
 - e.g. Reduce from 10^6 to 10^4
- For each class, find the top representative k features for it → get the Union over all classes → reduced feature space

Björn Ross, TTDS 2025/2026



23

Step 2: Feature Selection Functions

- Document frequency
 - % of docs in class c_i that contain the term t_k
 - Very basic measure. Will select stop words as features
$$\#(t_k, c_i) = P(t_k | c_i)$$
- Mutual Information
 - How much we learn from the presence or absence of term t_k about whether or not a document is in class c_i
 - Often used in feature selection in text classification
$$MI(t_k, c_i) = \sum_{c \in \{c_i, c_i\}} \sum_{t \in \{t_k, t_k\}} P(t, c) \cdot \log_2 \frac{P(t, c)}{P(t) \cdot P(c)}$$
- Pearson's Chi-squared (χ^2)
 - used more in comparisons between classes

Björn Ross, TTDS 2025/2026



24

12

原 PPT 第 12 页

人话直觉

如果 vocabulary 有百万维，很多列几乎全是 0。模型不仅算得慢，还可能把偶然噪声当规律。feature selection 是先筛掉没用列。

精确定义 / 机制：文本 feature vector 可达 10^6 维且非常 sparse，带来 high computational cost 和 overfitting。基本做法是为每个 class 找 top k representative features，再对所有 classes 取 union，形成 reduced feature space。Document frequency 只看 class 中出现比例，可能选出 stop words；mutual information 衡量 term presence/absence 对 class 的信息量；chi-squared 衡量 term appearance 与 class 是否独立以及偏离独立的程度。

$$\chi^2(t, c) = \sum_{i \in \{0,1\}} \sum_{j \in \{0,1\}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

考试问法：若问为什么 feature selection 重要，答 reduce dimensionality, computational cost, overfitting。若问方法，DF 有 stop-word 风险，MI/ χ^2 用 class association。

8. 8. Feature weighting: 同一个词出现了，也要问它有多重要

11/12/25

Step 2: Feature Selection Functions		
Function	Denoted by	Mathematical form
Document frequency	$\#(t_k, c_i)$	$P(t_k c_i)$
DIA association factor	$z(t_k, c_i)$	$P(c_i t_k)$
Information gain	$IG(t_k, c_i)$	$\sum_{c \in \{c_1, \dots, c_n\}} \sum_{t \in \{t_1, \dots, t_m\}} P(t, c) \cdot \log \frac{P(t, c)}{P(t) \cdot P(c)}$
Mutual information	$MI(t_k, c_i)$	$\log \frac{P(t_k, c_i)}{P(t_k) \cdot P(c_i)}$
Chi-square	$\chi^2(t_k, c_i)$	$\frac{[Tr] \cdot [P(t_k, c_i) \cdot P(\bar{t}_k, \bar{c}_i) - P(t_k, \bar{c}_i) \cdot P(\bar{t}_k, c_i)]^2}{P(t_k) \cdot P(\bar{t}_k) \cdot P(c_i) \cdot P(\bar{c}_i)}$
NGL coefficient	$NGL(t_k, c_i)$	$\sqrt{[Tr] \cdot [P(t_k, c_i) \cdot P(\bar{t}_k, \bar{c}_i) - P(t_k, \bar{c}_i) \cdot P(\bar{t}_k, c_i)]}$
Relevancy score	$RS(t_k, c_i)$	$\log \frac{P(t_k c_i) + d}{P(t_k \bar{c}_i) + d}$
Odds Ratio	$OR(t_k, c_i)$	$\frac{P(t_k c_i) \cdot (1 - P(t_k \bar{c}_i))}{(1 - P(t_k c_i)) \cdot P(t_k \bar{c}_i)}$
GSS coefficient	$GSS(t_k, c_i)$	$P(t_k, c_i) \cdot P(\bar{t}_k, \bar{c}_i) - P(t_k, \bar{c}_i) \cdot P(\bar{t}_k, c_i)$

Björn Ross, TTDS 2025/2026



25

Step 3: Feature Weighting	
- Attributing a value to feature t_k in document d_i. This value may be binary (representing presence/absence of t_k in d_i); numeric (representing the importance of t_k for d_i); obtained via feature weighting functions in the following two classes: unsupervised: e.g., tfidf or BM25, supervised: e.g., $tf \cdot MI$, $tf \cdot \chi^2$ - Similarity between two vectors may be computed e.g. via cosine similarity Scaling can be important!	

Björn Ross, TTDS 2025/2026



28

13

原 PPT 第 13 页

人话直觉

feature selection 决定“要不要这列”，feature weighting 决定“这列的数值是多少”。出现一次的 rare term 和出现一次的 stop word 不该同等对待。

精确定义 / 机制: Feature weighting 给 document 中的 feature t_k 赋值, 可为 binary presence/absence, 也可为 numeric importance, 如 term frequency、TF-IDF、BM25。Unsupervised weighting 不用 label, 如 TF-IDF/BM25; supervised weighting 可结合 label association, 如 $tf \cdot MI$ 或 $tf \cdot \chi^2$ 。Scaling 也重要, 因为不同特征量纲不同, 若不缩放可能让某些 features 不合理地主导模型。

$$w_{d,t} = tf_{d,t} \cdot idf_t$$

考试问法: 常见混淆: selection 是挑 feature, weighting 是给 feature value。答题时可举 binary、tf、TF-IDF、BM25、 $tf \cdot MI$ 。

9.9. Classifiers: 没有一种算法通吃所有文本任务

11/12/25

Training a Classifier

- For **Multiclass classification**, some learning algorithms for binary classification are "SLMC-ready"; e.g.
 - Decision trees
 - Random forests
 - Naive Bayesian methods
 - Lazy learning methods (e.g., k-NN)
 - Neural networks
- For other learners (notably: SVMs) to be used for SLMC classification, combinations / cascades of the binary versions need to be used
 - e.g. multi-class classification SVM
 - Could be directly used for MLMC as well

Björn Ross, TTDS 2025/2026

THE UNIVERSITY of EDINBURGH

31

Word embeddings

- More complex representation of words as vectors

Recall the **term vectors** in traditional indexing or classification, using term frequency as weights:

- Sparse (most values are 0)
- Capture semantics only incidentally (similar vectors are terms that appear together)

	he	drink	ink	likes	pink	think	wink
2	1	0	2	0	0	1	
1	3	0	1	0	0	0	
1	1	1	1	0	1	0	
1	1	1	1	1	0	0	
1	1	1	1	1	0	1	

← D1: He likes to wink, he likes to drink
← D2: He likes to drink, and drink, and drink
← D3: The thing he likes to drink is ink
← D4: The ink he likes to drink is pink
← D5: He likes to wink, and drink pink ink

Björn Ross, TTDS 2025/2026

THE UNIVERSITY of EDINBURGH

32

15

原 PPT 第 15 页

人话直觉

分类器像不同工具：SVM、Naive Bayes、k-NN、random forest 都能干活，但没有一把瑞士军刀永远最好。

精确定义 / 机制： Classical binary classifiers 包括 SVM、random forests、Naive Bayes、k-NN、logistic regression 等。No-free-lunch principle 表示没有一种 learning algorithm 在所有 contexts 中都优于其他。文本分类实现要适应 high-dimensional sparse representation。某些算法天然支持 multi-class，如 decision trees、random forests、Naive Bayes、k-NN、neural networks；SVM 做 multi-class 通常需要组合或级联 binary SVM，如 one-vs-rest。

考试问法： 若问 algorithm choice，答 no-free-lunch and task/data-dependent。若问 SVM multi-class，说明需要 binary decomposition/cascade。

10. 10. Embeddings and PLMs: 从稀疏词表到上下文语义

11/12/25

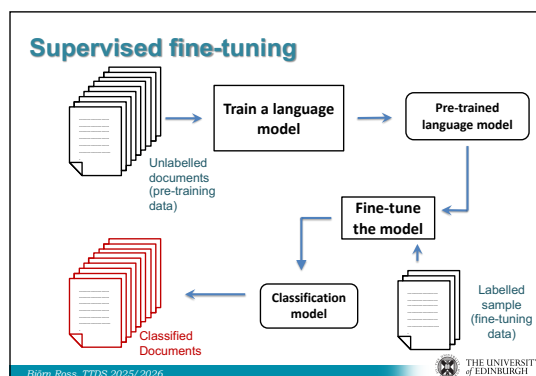
Pre-trained language models

- Obtained through self-supervised learning
 - Predict the next word: The Queen of [...] (*next token prediction*)
 - Predict a masked word: The [...] of England. (*masked language modelling*)
 - ...
- Also done on large unlabelled pre-training corpora
- Penultimate layer of network can be used to generate **contextualised word embeddings** for other language-based tasks
- Basis for many state-of-the-art text classifiers
 - e.g. BERT (DistilBERT, RoBERTa...), XLNet, etc.

Bylrm, Ross, TTDS 2025/2026

THE UNIVERSITY of EDINBURGH

35



36

17

原 PPT 第 17 页

人话直觉

BOW 里 cat 和 kitten 是两列互不认识的 0/1; embedding 让相似词在向量空间里靠近; PLM 进一步让 bank 在 river bank 和 bank account 中有不同表示。

精确定义 / 机制: Word embeddings 是 dense vectors, 通常所有 entries 非零, 能更直接捕捉 semantic similarity; 可用 self-supervised learning 训练, 如 CBOW 根据上下文预测中心词, skip-gram 根据目标词预测上下文。Pre-trained language models 在大规模未标注语料上做 next token prediction 或 masked language modelling, 产生 contextualized embeddings, 同一词在不同上下文中可有不同表示。Supervised fine-tuning 是先预训练 LM, 再用 labeled task data 微调成 classification model。

考试问法: 比较题: BOW sparse and indirect semantics; embeddings dense semantic vectors; PLMs contextual and pre-trained on large unlabelled corpora。

11. 11. Zero-shot and few-shot: 用 prompt 临时搭一个分类器

11/12/25

Zero-shot classification

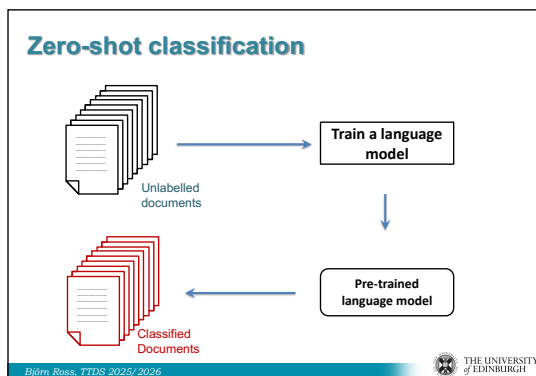
- Using language models for classification directly without giving training examples (= skipping the fine-tuning step)
 - A modern approach that can work well and requires little human effort
 - Depends highly on black-box models, limited opportunities for customisation and error analysis (but this is an active research area)
 - Often used as a baseline for performance comparisons

You
Consider the following sentence: "I love the book." What sentiment is it? Your reply must be one of the following words: positive, neutral, or negative.

ChatGPT
Positive.

Björn Ross, TTDS 2025/2026

37



38

18

原 PPT 第 18 页

人话直觉

zero-shot 像直接问一个大模型“这条评论是正面还是负面?”; few-shot 则先给它看几道样题, 再让它接着做。

精确定义 / 机制: Zero-shot classification 不提供 task-specific training examples, 也跳过 fine-tuning, 直接用 language model 按 prompt 分类。优点是人工成本低, 能快速建立 baseline; 缺点是依赖 black-box model, customisation 和 error analysis 空间有限。Few-shot classification 在 prompt 中给少量 examples, 不做参数 fine-tuning, 利用 in-context learning 适配任务。

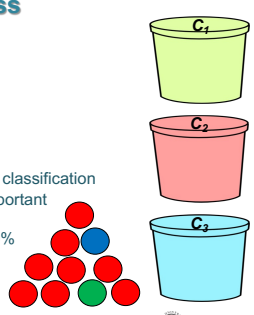
考试问法: 若问 zero-shot/few-shot 区别: few-shot has examples in prompt; 二者都 skip task-specific fine-tuning。评价时可作为 baseline, 但不要把它当成无须验证的最终模型。

12. 12. Evaluation: accuracy 很诱人，但经常骗人

11/12/25

Evaluation: Multi-class

- Majority class baseline
 - Accuracy = 0.8
 - Macro-F1 = 0.296
- Macro-F1:
 - Should be used in binary classification when two classes are important
 - e.g.: males/females while distribution is 80/20%



Bylrm Ross, TTDS 2025/2026

THE UNIVERSITY of EDINBURGH

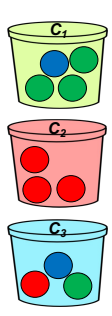
47

Error Analysis

- Confusion Matrix

	Predicted class		
	Green	Red	Blue
Actual class			
Green	3	0	1
Red	0	3	1
Blue	1	0	1

- Useful:
 - Find classes that are confused with others
 - Develop better features to solve the problem



Bylrm Ross, TTDS 2025/2026

THE UNIVERSITY of EDINBURGH

48

23

原 PPT 第 23 页

人话直觉

如果 80% 样本都是 C1，一个永远猜 C1 的笨模型 accuracy 也有 0.8；但它对 C2/C3 完全不会。

精确定义 / 机制： Evaluation 包括 effectiveness、efficiency 和 baselines. Effectiveness measures 有 accuracy、precision、recall、F1、per-class metrics、macro-F1；efficiency 可看 training speed、classification speed、feature extraction speed；baselines 包括 random、majority class、simple BOW、LLM zero-shot. Accuracy 在 classes 平衡且同等重要时合适；imbalanced data 中可能被 majority class baseline 掩盖。Macro-F1 先算每个 class 的 F1 再平均，给 minority classes 同等权重。Data splitting 用 training/test 或 training/development/test；development set 用于 hyperparameter optimisation, test set 只能做最终评估。

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{MacroF1} = \frac{1}{|C|} \sum_{c \in C} F1_c$$

考试问法： 若考 imbalanced classification，明确说 accuracy misleading；应报告 minority/each-class

precision, recall, F1 and macro-F1, 并与 majority baseline 比较。若问调参, 强调在 test set 上调参是 cheating。

考试速记版

- Classification = predict predefined finite classes; clustering = groups unknown。
- General mapping: $h:D \rightarrow C$; MLMC: $h:D \rightarrow 2^C$ 。
- Binary/SLMC/MLMC 要会定义、举例、写公式。
- Rule-based: interpretable/no labels but expensive/low recall。
- Supervised pipeline: label $\rightarrow$ features $\rightarrow$ train $\rightarrow$ apply/evaluate。
- BOW feature size = vocabulary size; preprocessing is task-dependent。
- Feature selection reduces high-dimensional sparse overfitting; MI/ χ^2 use label association。
- Feature weighting: binary/tf/TF-IDF/BM25/tf*MI/tf* χ^2 。
- Embeddings are dense semantic; PLMs provide contextual representations。
- Zero-shot/few-shot skip fine-tuning; useful baselines but need evaluation。
- Imbalanced data: accuracy can mislead; use per-class P/R/F1 and macro-F1。
- Never tune hyperparameters on test set; use development set。

一段稳妥考场回答

如果考 text classification pipeline, 可以这样答: text classification 是学习一个 classifier h , 把文本数据项 D 映射到预定义有限类别 C 。实际 supervised pipeline 先人工标注 representative samples, 再把文本转换为 numerical feature vectors, 例如 BOW、n-grams、syntactic/topic/non-textual features; 由于文本表示高维稀疏, 通常需要 feature selection 降低计算成本和 overfitting, 并用 binary、tf、TF-IDF、BM25 或 supervised weights 给 features 赋值。随后训练合适 classifier, 如 Naive Bayes、SVM、logistic regression、random forest 或 neural model; 现代方法还可用 embeddings、pre-trained language models、zero-shot/few-shot prompts。最后必须在 unseen data 上评估, 使用 accuracy、precision、recall、F1、macro-F1 等指标, 并与 baselines 比较; 若 classes imbalanced, accuracy 可能被 majority class baseline 误导, 应重点报告 per-class 和 macro-F1。